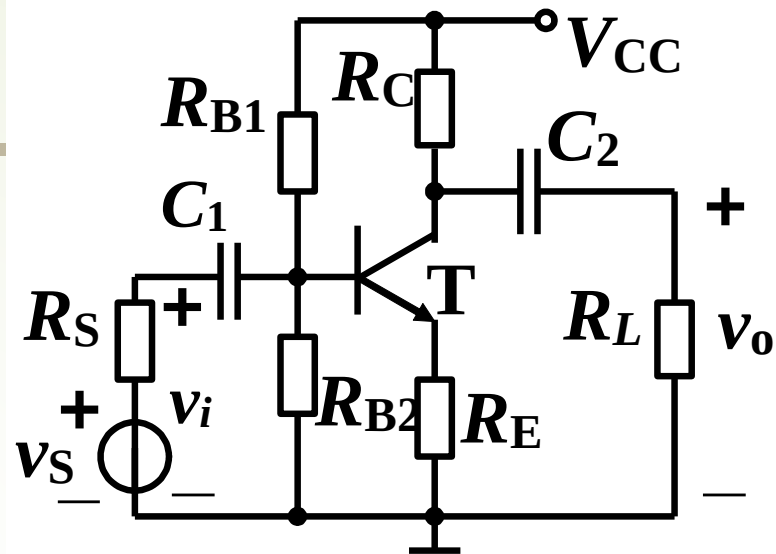


分析举例 (续)

例2 (1) 求Q点

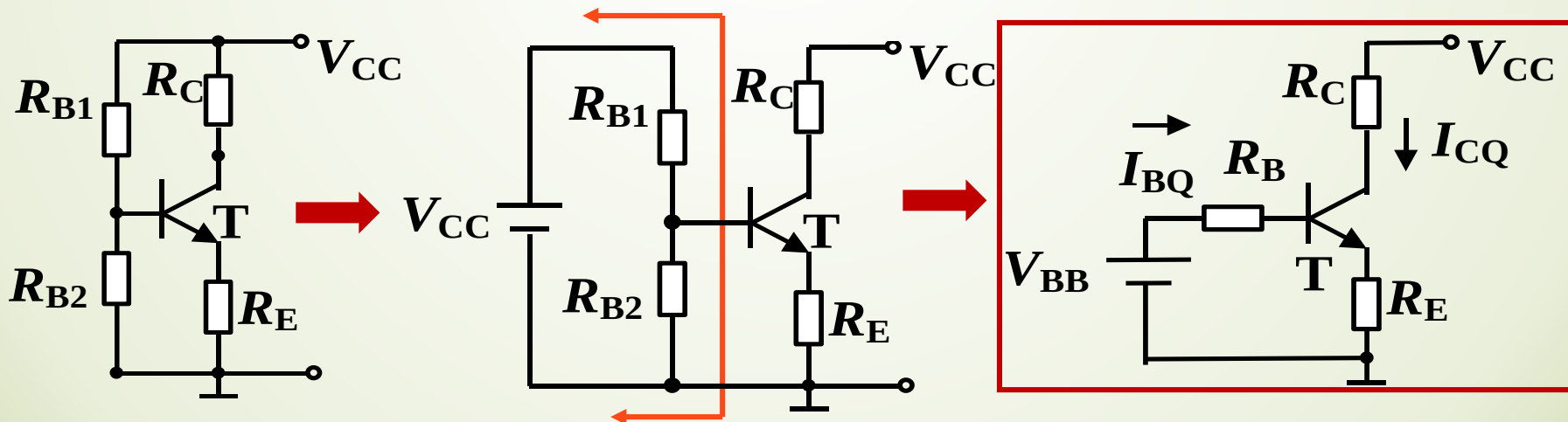
等效电压源

$$\begin{cases} V_{BB} = \frac{R_{B2}}{R_{B1} + R_{B2}} V_{CC} \\ R_B = R_{B1} // R_{B2} \end{cases}$$



$$I_{BQ} = \frac{V_{BB} - V_{BEQ}}{R_B + (\beta + 1)R_E}$$

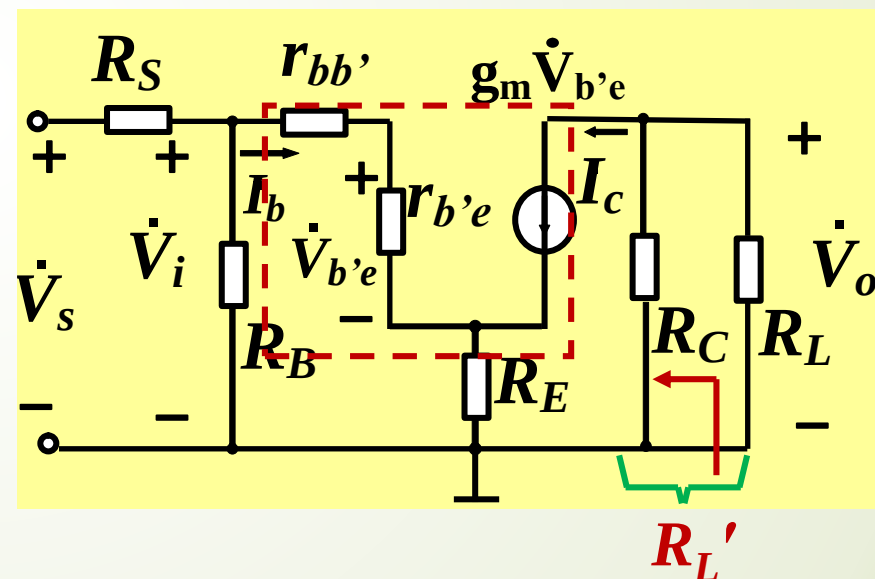
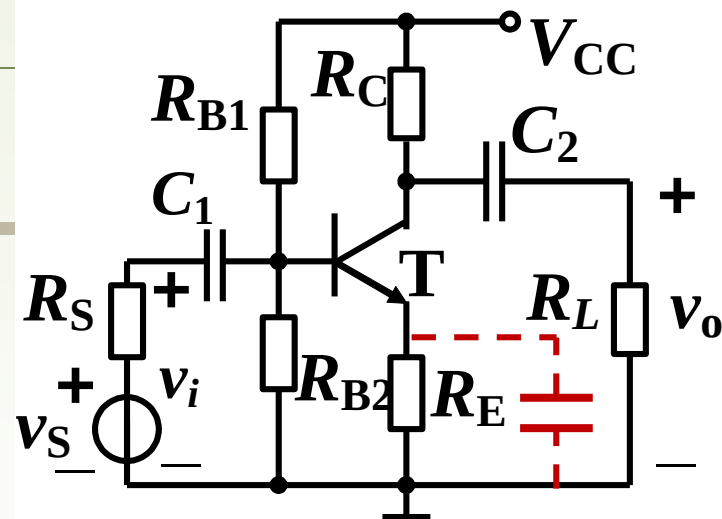
$$\begin{aligned} I_{CQ} &= \beta I_{BQ} \\ V_{CEQ} &\approx V_{CC} - (R_C + R_E)I_{CQ} \end{aligned}$$



R_E 作用：
负反馈，稳定工作点。 $A_v \downarrow$
并联C可避免 $A_v \downarrow$ 。

例 2 (2) 动态参数

交流等效电路，忽略 r_{ce}



由于忽略 r_{ce} ，理想电流源内阻无穷大，可忽略 R_E

$$\begin{cases} \dot{V}_i = (r_{bb'} + r_{b'e})\dot{I}_b + (\dot{I}_b + g_m \dot{V}_{b'e})R_E \\ \dot{V}_{b'e} = \dot{I}_b r_{b'e} \\ \dot{V}_o = -g_m \dot{V}_{b'e} R_L' \end{cases}$$

$$\rightarrow A_v = \frac{\dot{V}_o}{\dot{V}_i} = \frac{-\beta R_L'}{r_{be} + (\beta + 1)R_E}$$

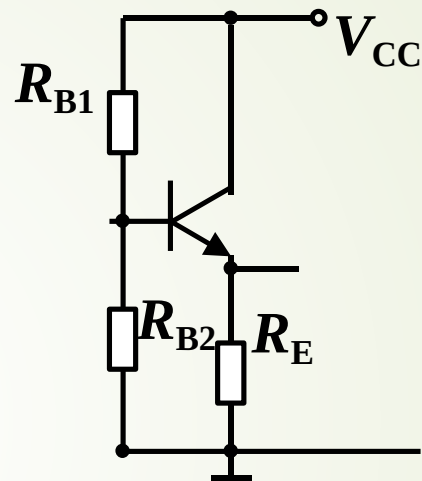
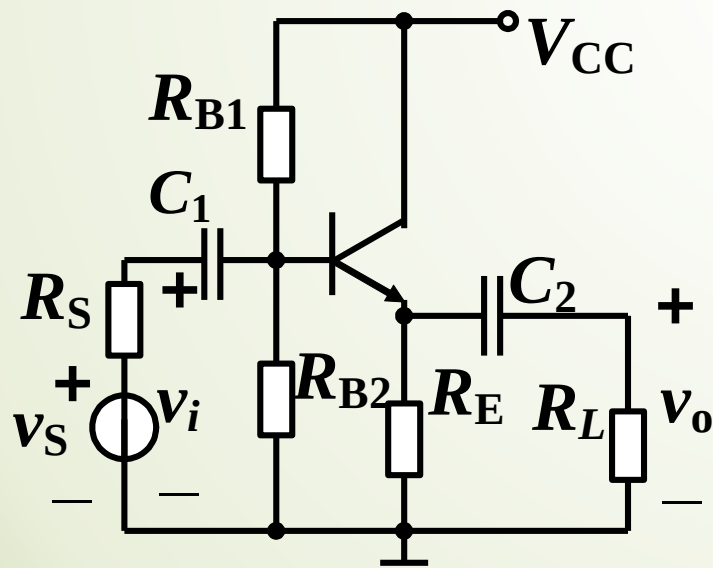
$$R_i = \frac{V_i}{I_i} = R_B // [r_{be} + (\beta + 1)R_E]$$

$$R_o \approx R_C$$

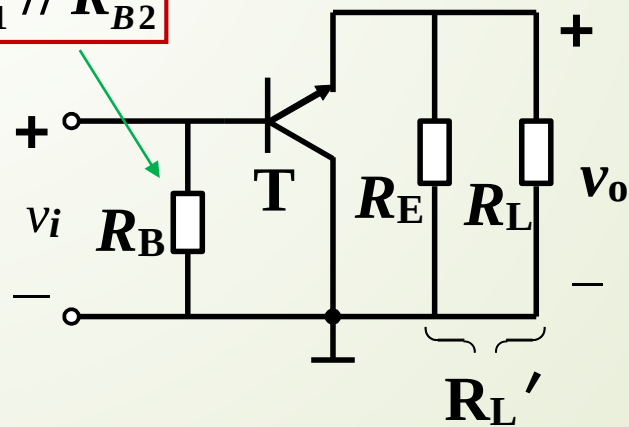
共集放大电路 (CC)

CC电路的基本特性

1. 电路构成
2. 直流通路
3. 交流通路

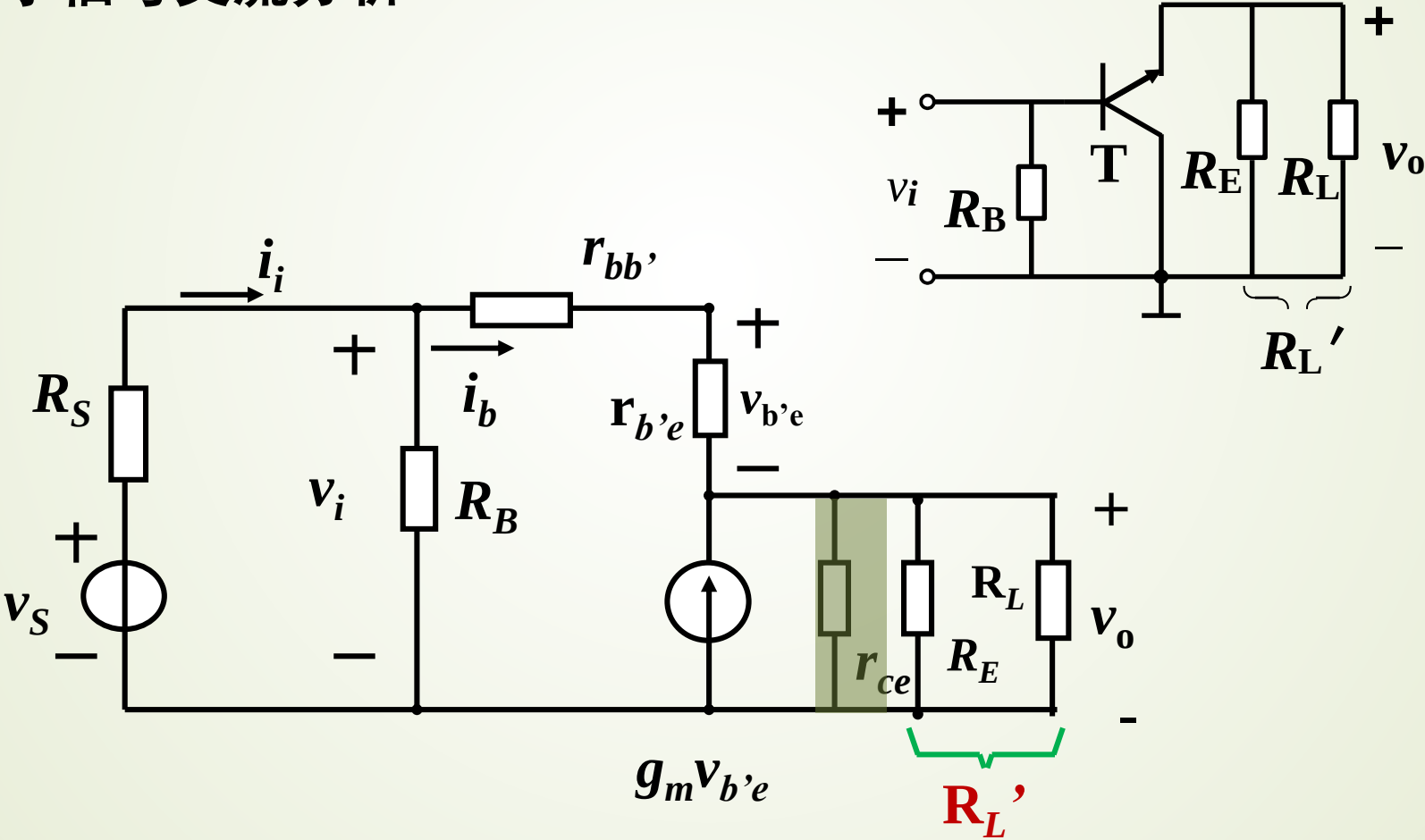


$$R_B = R_{B1} // R_{B2}$$



CC电路的基本特性（续）

4. 小信号交流分析

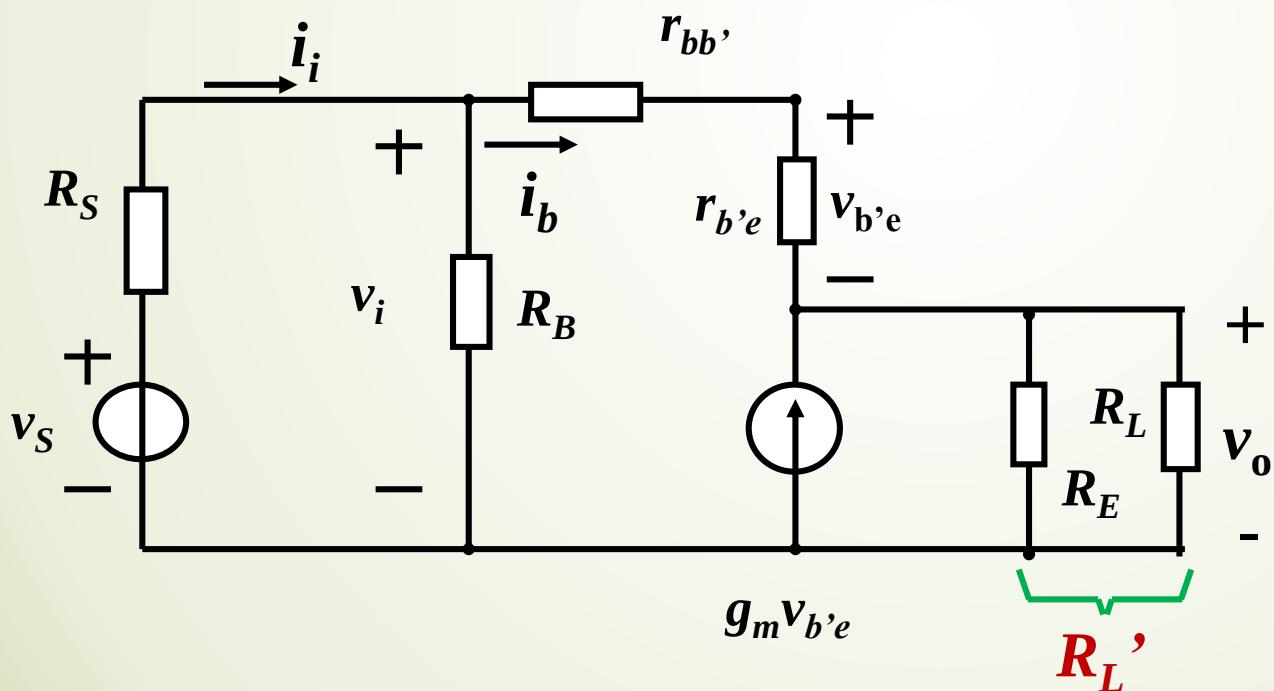


CC电路的基本特性（续）

4. 小信号交流分析

$$(1) A_v = \frac{v_o}{v_i}$$

$$\begin{cases} v_i = r_{be} i_b + v_o \\ v_o = (i_b + g_m v_{b'e}) R'_L = (1 + g_m r_{b'e}) i_b R'_L \\ = (\beta + 1) i_b R'_L \end{cases}$$



$$\dot{A}_v = \frac{(\beta + 1) R'_L}{r_{be} + (\beta + 1) R'_L}$$

$< 1, \rightarrow 1$ 同相

射极跟随器

CC电路的基本特性（续）

(2) R_i

$$R_i = \frac{v_i}{i_i} = R_B // R'_i$$

$$R'_i = \frac{v_i}{i_b} = r_{be} + (\beta + 1)R'_L$$

$$R_i = R_B // [r_{be} + (\beta + 1)R'_L]$$

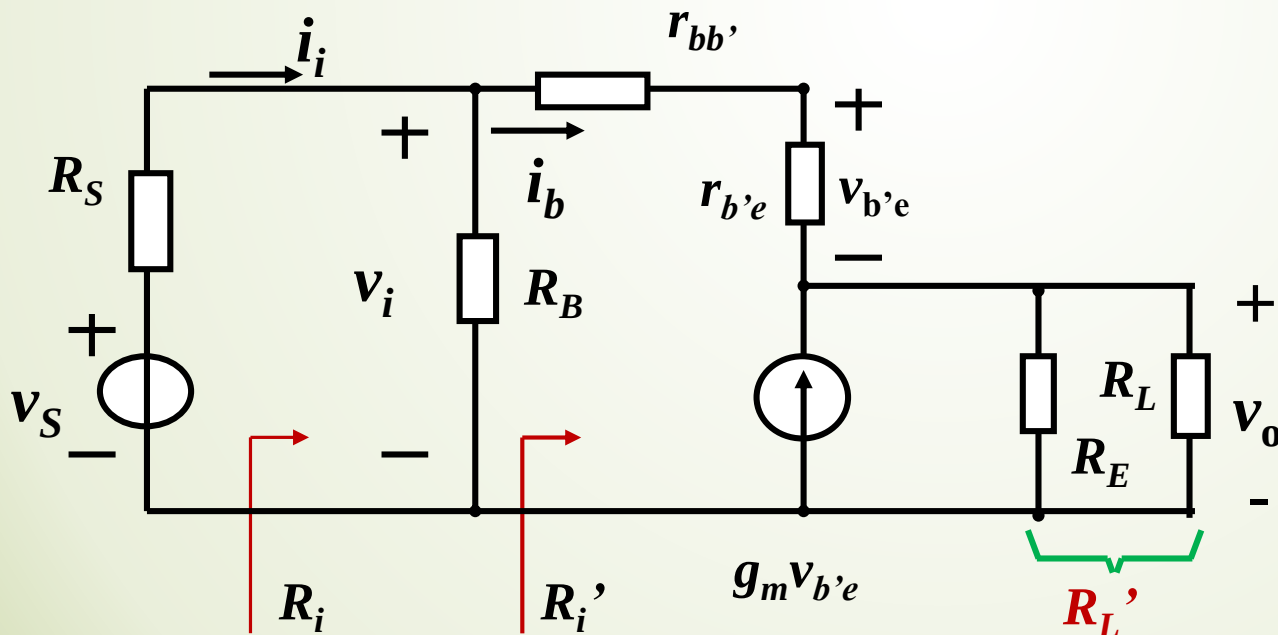
R'_i 大!

(不考虑 R_B 的影响时)

基极偏置 R_{B1}, R_{B2}

宜大不宜小!

R_i 与 R'_L 有关!



CC电路的基本特性 (续)

(3) R_o

$$R_o = \frac{V}{I} = R_E // R_o'$$

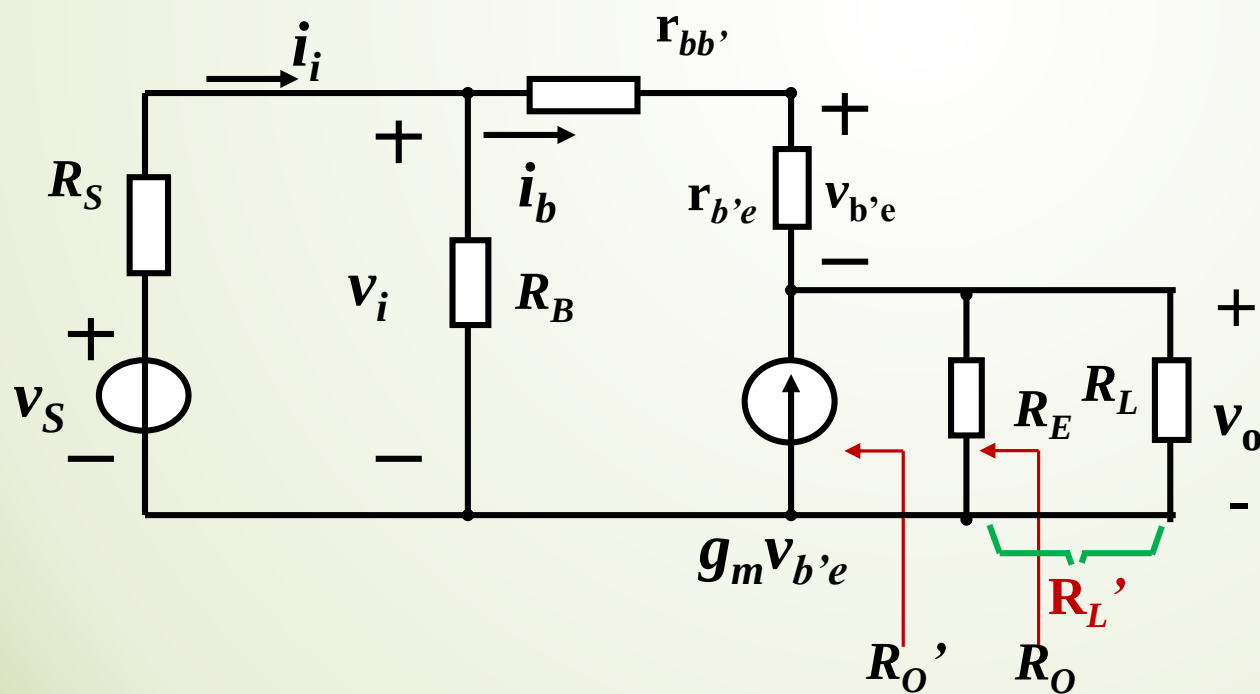
$$R_o' = \frac{V}{I'} = - \frac{V}{(I_b + g_m r_{b'e} i_b)} = \frac{1}{\beta + 1} \bullet \frac{V}{-i_b} = \frac{R_S' + r_{be}}{\beta + 1}$$

$$R_S' = R_S // R_B$$

$$R_o = R_E // \frac{R_S' + r_{be}}{\beta + 1}$$

R_o 小!

R_o 与 R_S 有关!



共集放大电路的主要特点及应用

$$\dot{A}_v = \frac{(\beta + 1)R'_L}{r_{be} + (\beta + 1)R'_L}$$

$$R'_i = r_{be} + (\beta + 1)R'_L$$

$$R_O = R_E // \frac{R'_S + r_{be}}{\beta + 1}$$

特点：

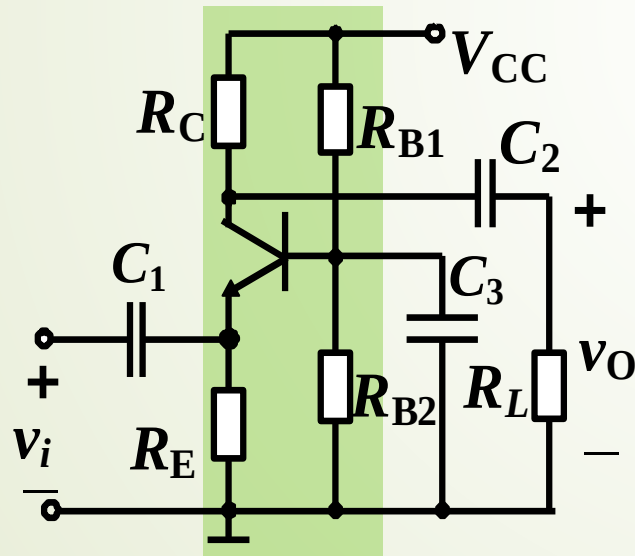
1. 电压增益接近 “1”
2. 高输入阻抗、低输出阻抗

应用：

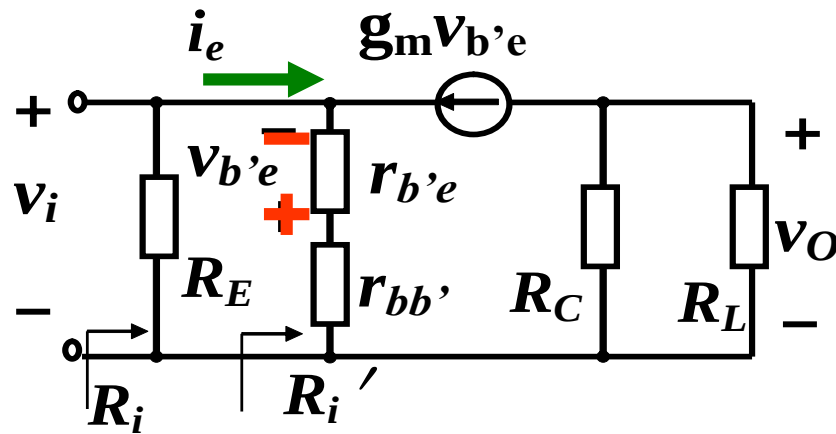
1. 射极跟随器，电平移动
2. 阻抗变换器：
作为多级放大器的输入级，输出级或中间缓冲级

共基放大电路 (CB)

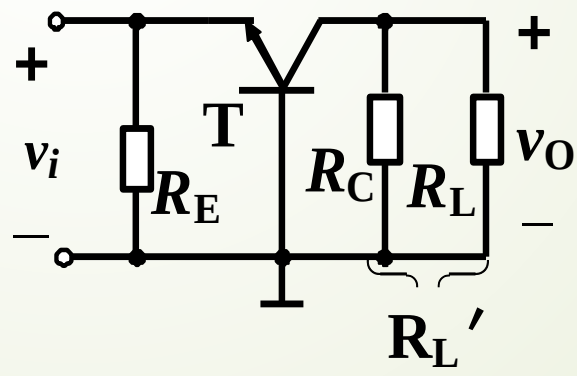
电路组成



CB 放大电路及其交流通路



CB 交流等效电路



共基放大电路 (续)

$$(1) A_v = \frac{v_o}{v_i}$$

$$v_o = -g_m v_{b'e} R'_L$$

$$= -g_m R'_L \frac{-r_{b'e}}{r_{be}} v_i$$

$$A_v = \frac{\beta R'_L}{r_{be}}$$

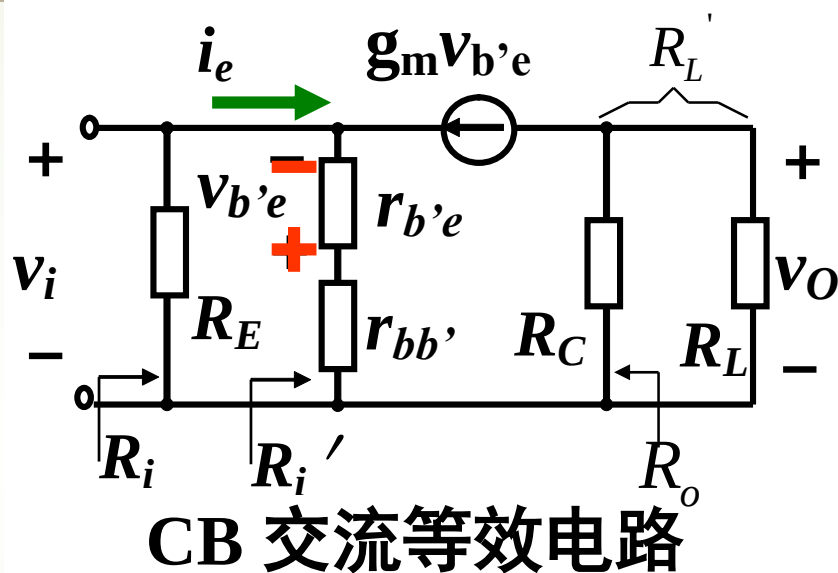
同相!
大小与
CE相同

$$(2) R_i$$

$$R_i = R_E // R'_i$$

$$R'_i = \frac{v_i}{i_e} = \frac{V_{be}}{(\beta + 1)i_b} = \frac{r_{be}}{\beta + 1}$$

$$R_i = R_E // \frac{r_{be}}{\beta + 1} \approx r_e = \frac{V_T}{I_{EQ}}$$



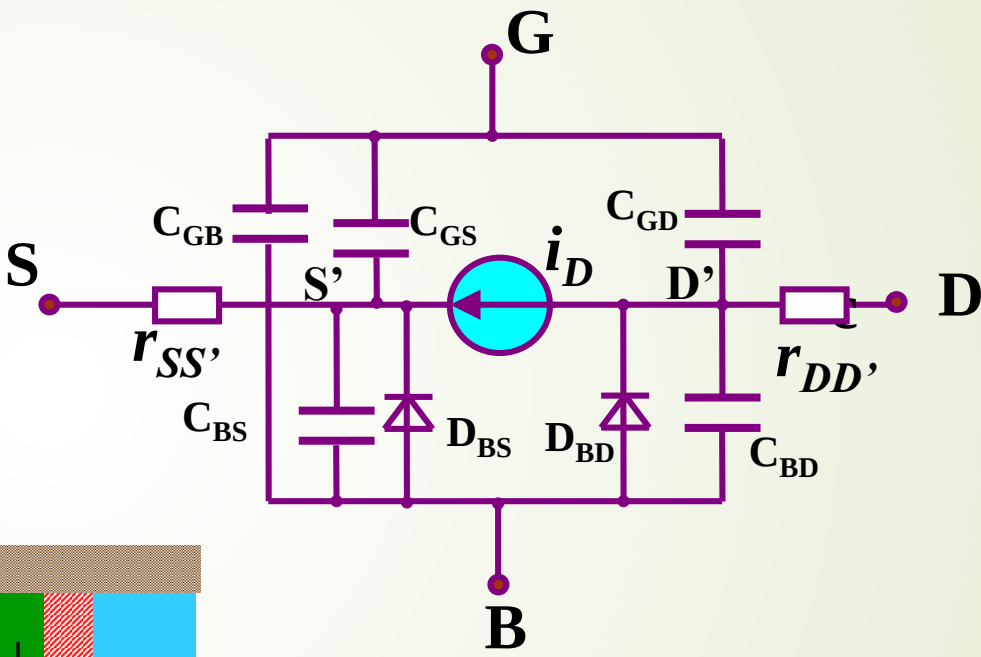
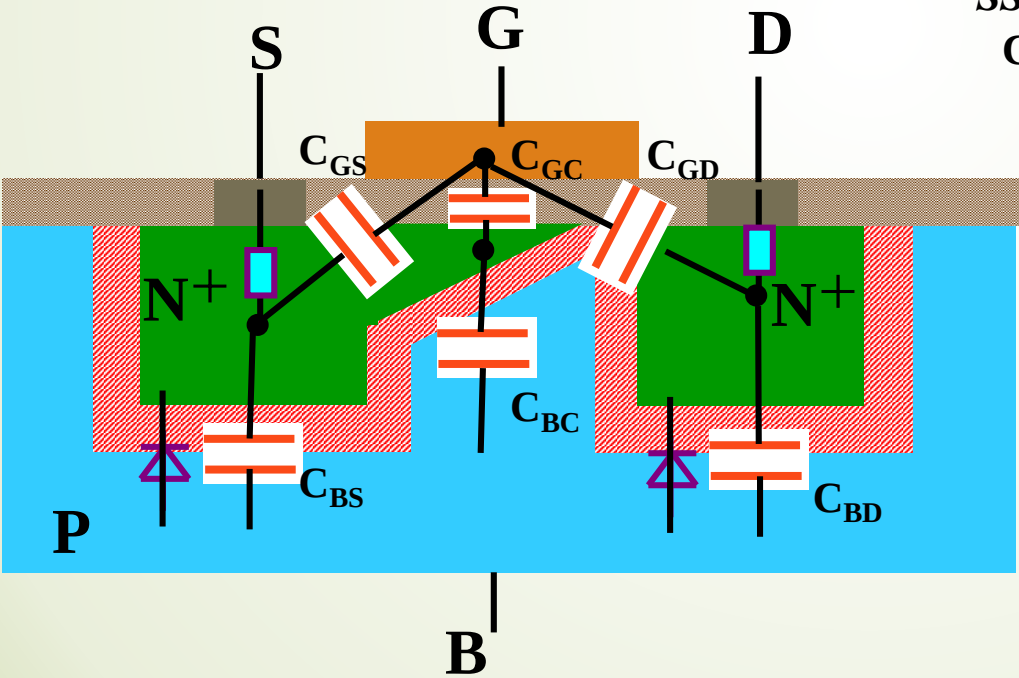
$$(3) R_o$$

$$R_o \approx R_C$$

特点:

A_v + 同相放大
 R_i 低

MOSFET的模型

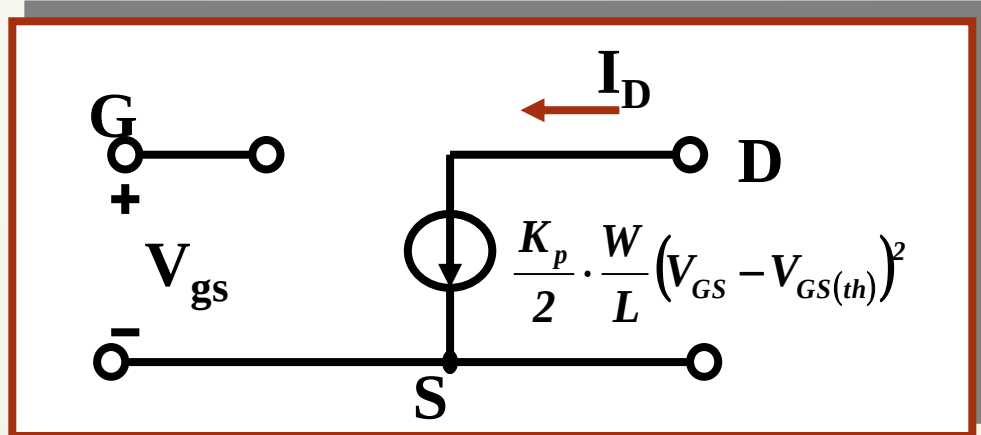


MOS管的模型

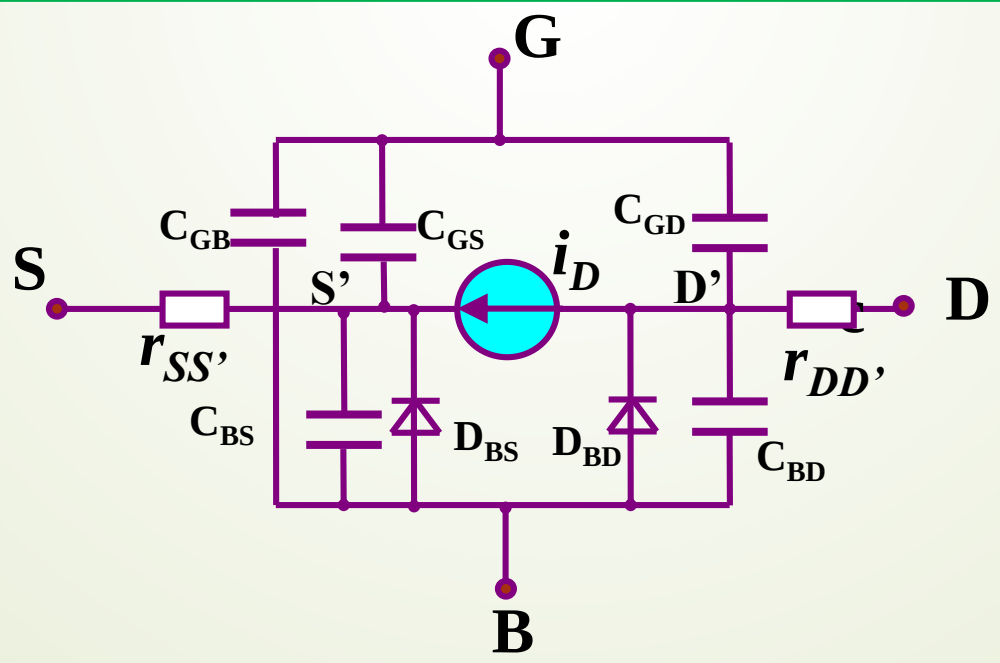
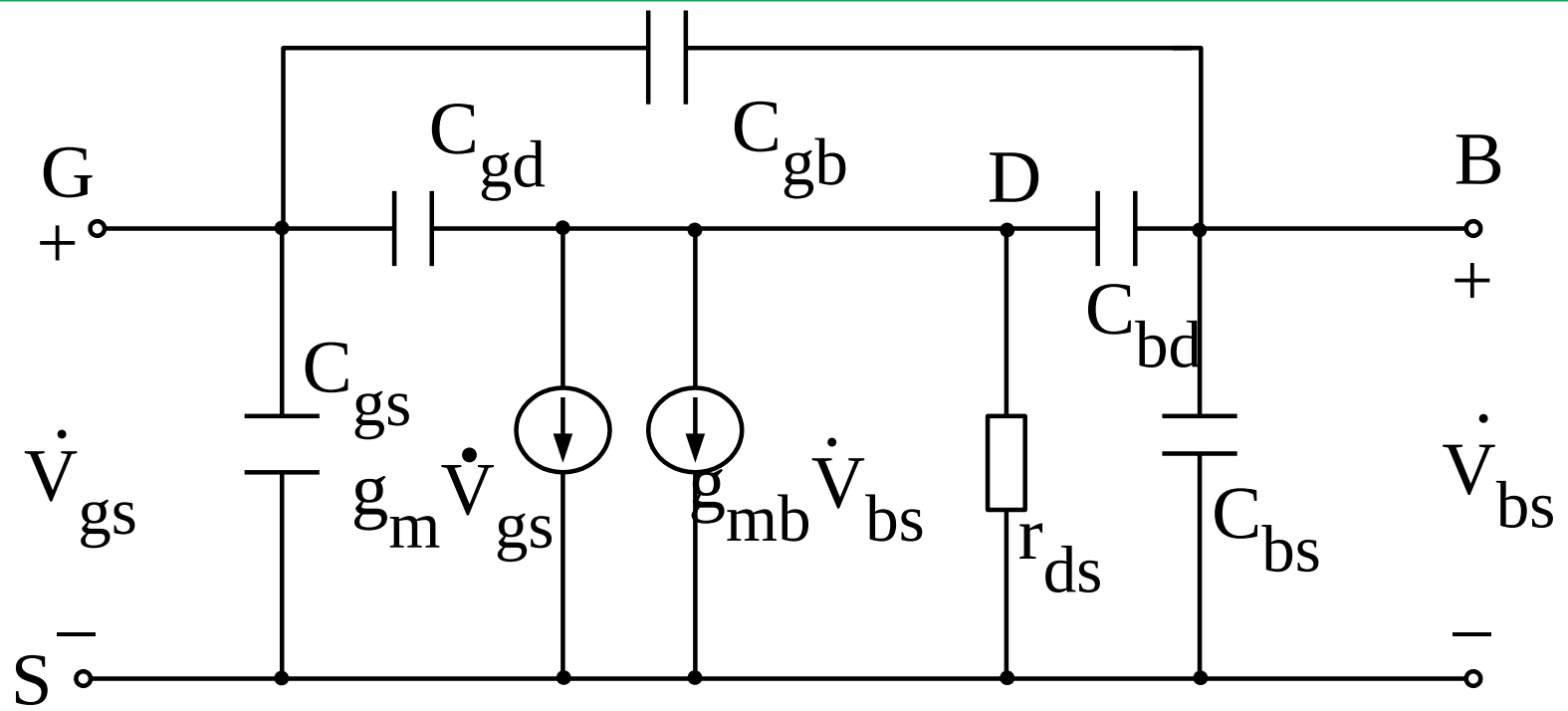
1. 简化直流模型

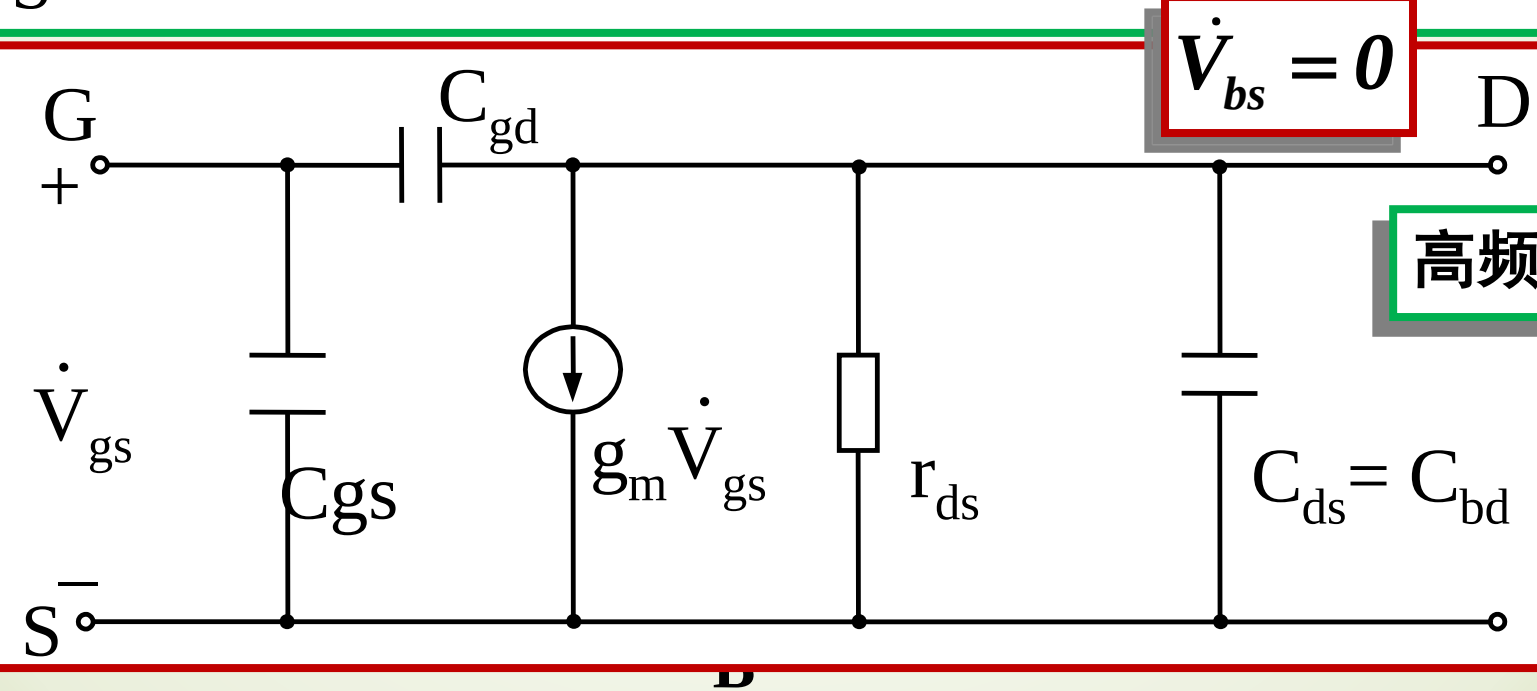
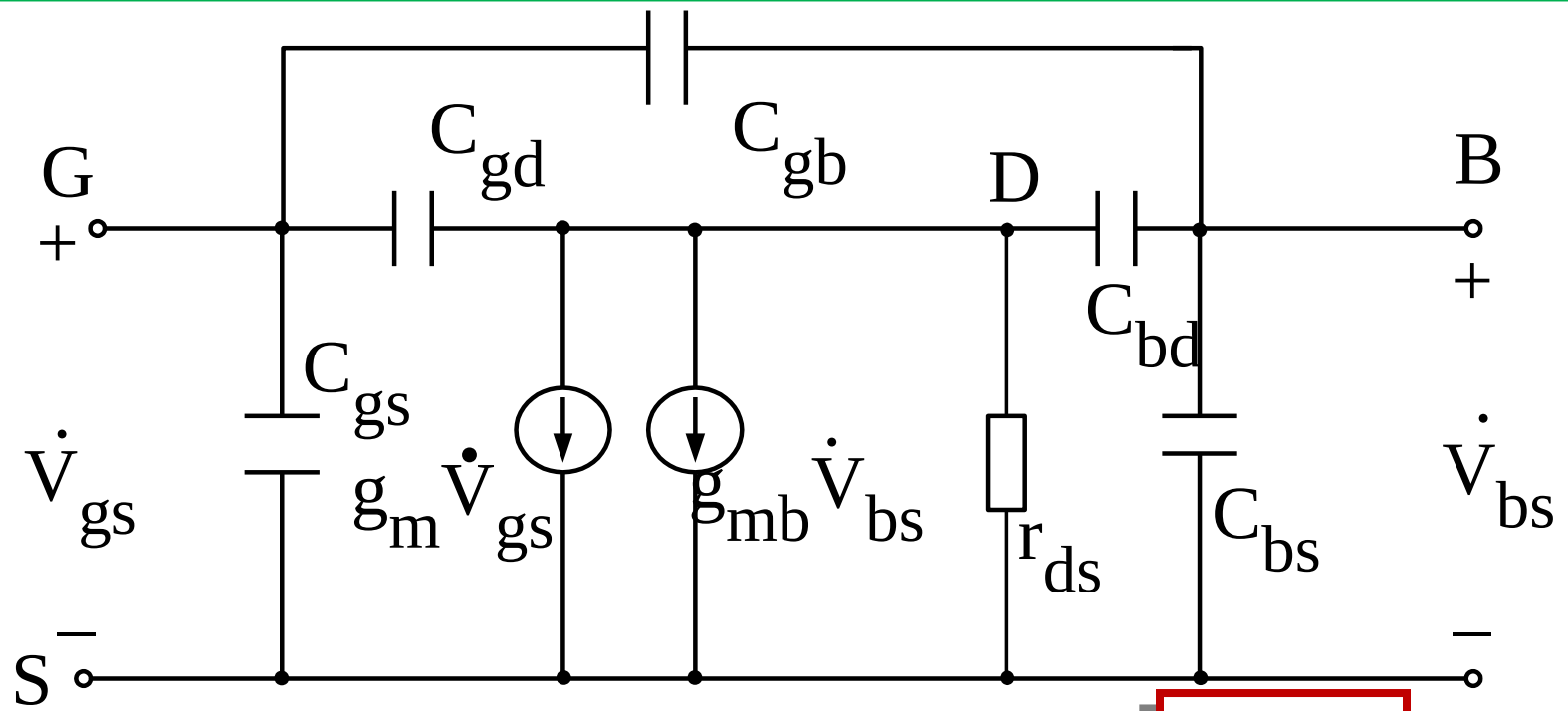
$$\begin{cases} I_G = 0 \\ I_D \approx \frac{K_p}{2} \cdot \frac{W}{L} (V_{GS} - V_{GS(th)})^2 \end{cases}$$

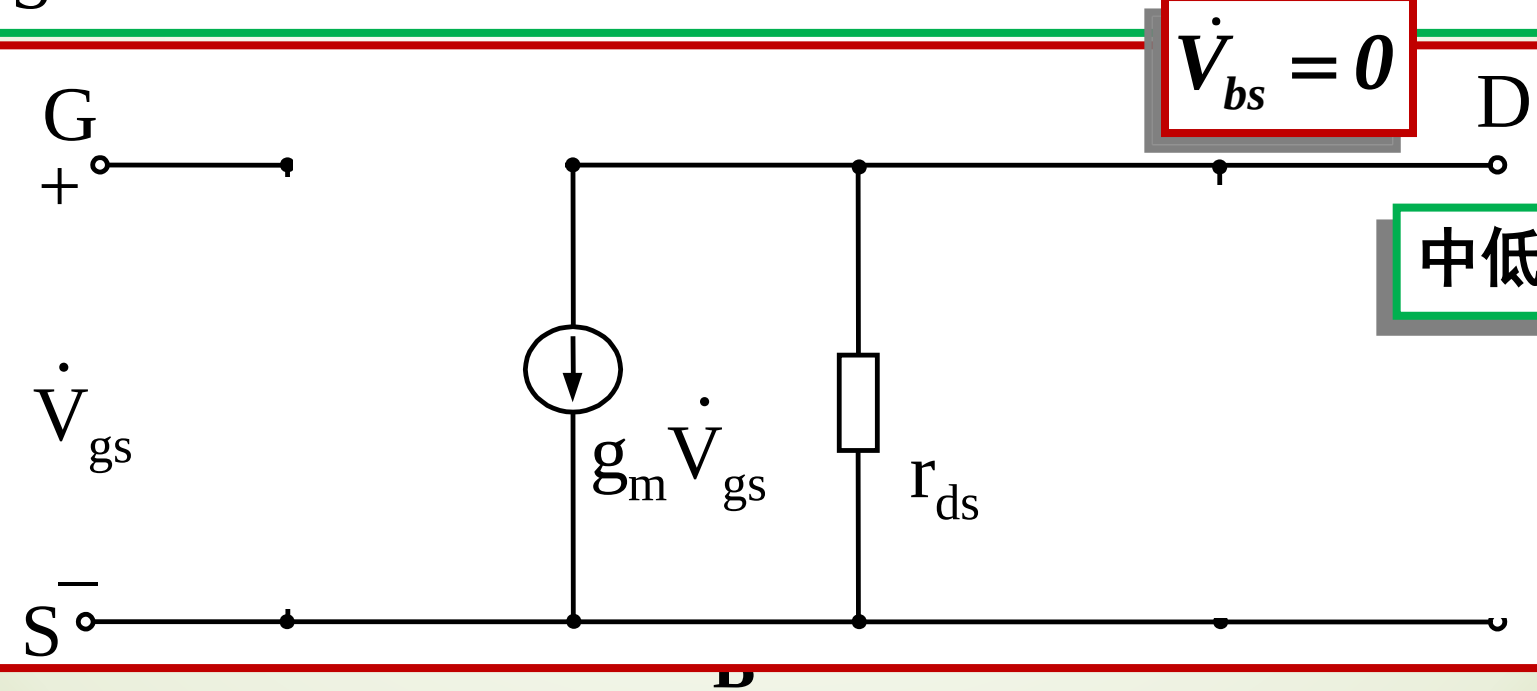
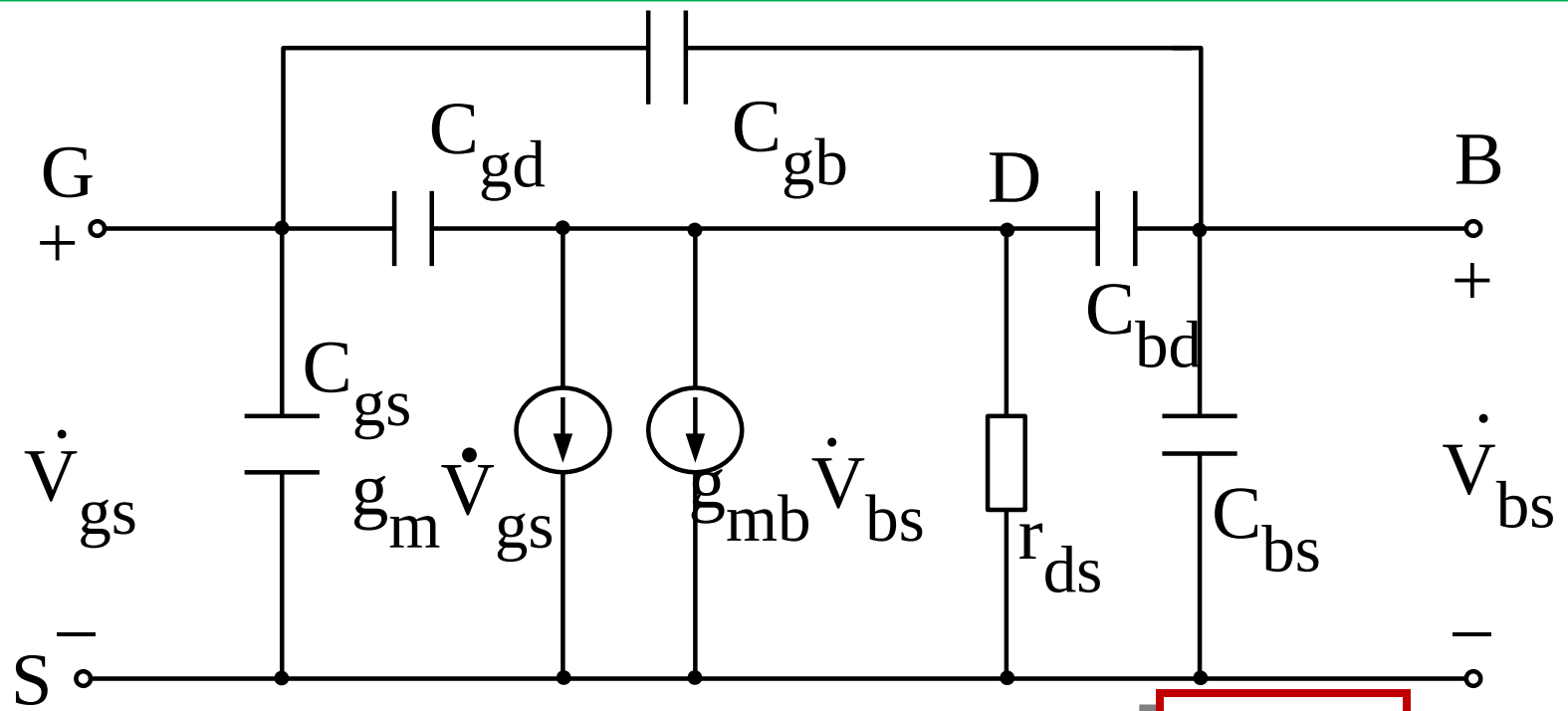
$K_p = \mu_n C_{ox}$
本征导电因子



2. 交流小信号模型







MOS管的模型 (续)

$$i_D = \frac{k_p}{2} \frac{W}{L} (V_{GS} - V_{GS(th)})^2 (1 + \lambda v_{DS})$$

$$g_m = k_p \cdot \frac{W}{L} (V_{GS} - V_{GS(th)}) (1 + \lambda v_{DS})$$

$$g_m = \frac{2i_D}{V_{GS} - V_{GS(th)}}$$

$$g_m = \sqrt{2k_p \cdot \frac{W}{L} (1 + \lambda v_{DS}) i_D}$$

$$\frac{1}{r_{ds}} = g_{ds} = \frac{k_p}{2} \cdot \frac{W}{L} (V_{GS} - V_{GS(th)})^2 \lambda = \frac{\lambda I_D}{1 + \lambda V_{DS}}$$

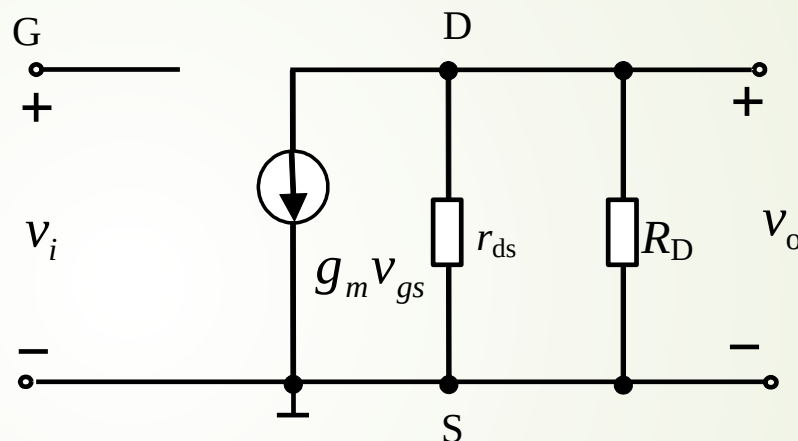
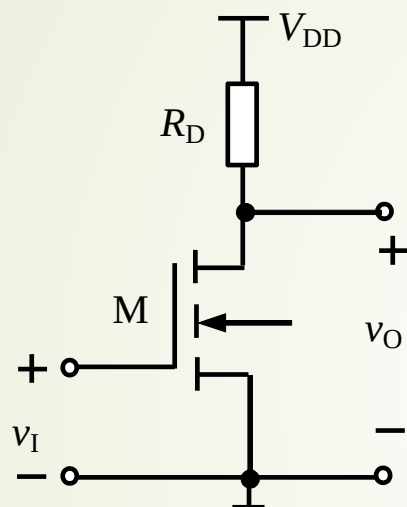
对 v_{DS} 求导

$$r_{ds} = \frac{1 + \lambda V_{DS}}{\lambda I_D} = \frac{V_A + V_{DS}}{I_D} \approx \frac{V_A}{I_D}$$

沟道长度调制系数:

$$\lambda = \frac{1}{V_A}$$

基本共源极 (CS) 放大电路



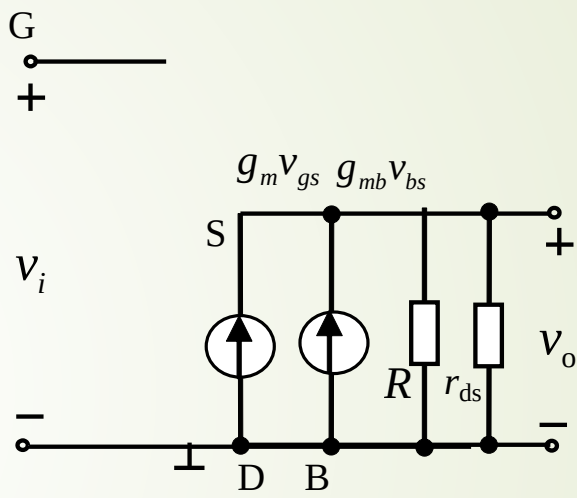
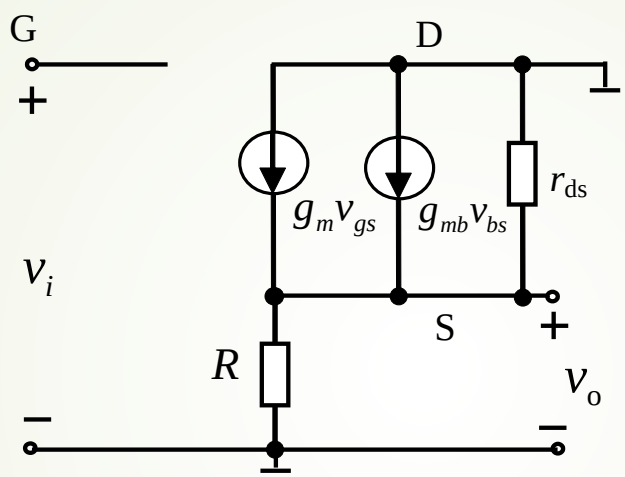
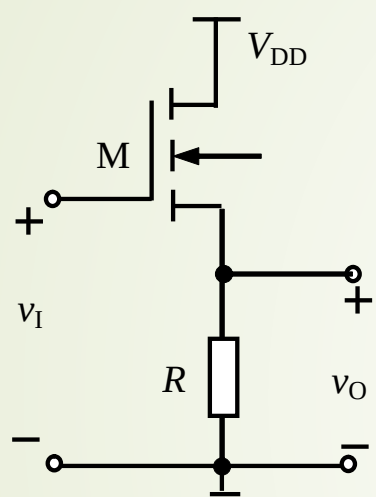
$$I_D \approx \frac{k_p}{2} \frac{W}{L} (V_{GS} - V_{GS(th)})^2$$

$$V_{DSQ} = V_{DD} - R_D I_{DQ}$$

$$\dot{A}_v = -g_m (R_D // r_{ds})$$

$$R_o = R_D // r_{ds}$$

共漏极（CD）放大电路



$$R'_L = R // r_{ds}$$

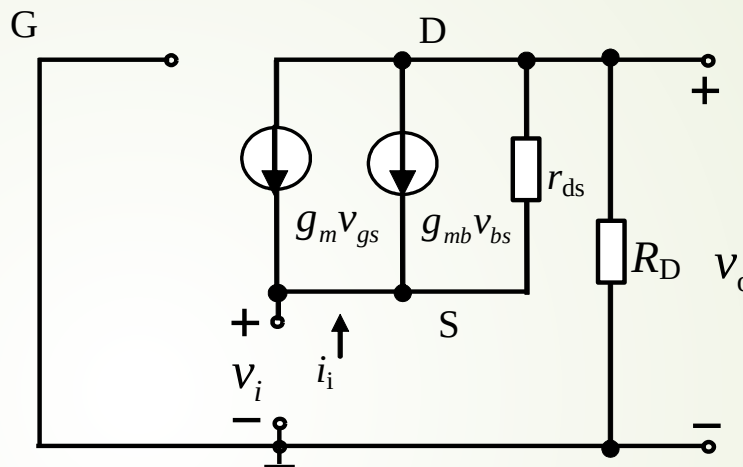
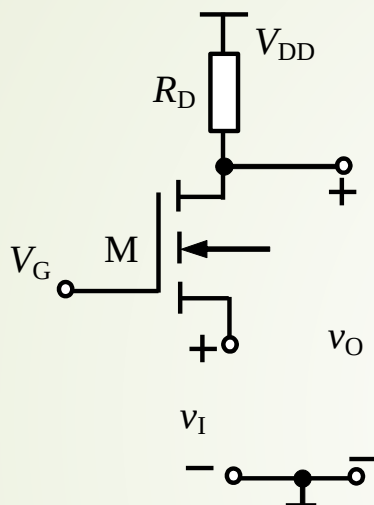
$$\begin{cases} v_{gs} = v_i - v_o \\ v_{bs} = -v_o \\ v_o = (g_m v_{gs} + g_{mb} v_{bs}) R'_L \end{cases}$$

$$A_v = \frac{g_m R'_L}{1 + (g_m + g_{mb}) R'_L}$$

$$R_o = \frac{1}{g_m} // \frac{1}{g_{mb}} // r_{ds} // R = \frac{1}{g_m + g_{mb} + \frac{1}{r_{ds}} + \frac{1}{R}}$$

源极跟随器

共栅极 (CG) 放大电路



$$v_o = [(g_m + g_{mb})v_i + (v_i - v_o)/r_{ds}]R_D$$

$$R_o = R_D // r_{ds}$$

$$A_v = \frac{(g_m + g_{mb} + 1/r_{ds})R_D}{1 + R_D/r_{ds}}$$

$$\begin{cases} i_i = v_i(g_m + g_{mb}) + (v_i - v_o)/r_{ds} \\ v_o = A_v v_i \end{cases}$$

当 $r_{ds} \gg R_D$ 时,

$$A_v \approx (g_m + g_{mb})R_D$$

$$R_i = \frac{v_i}{i_i} = \frac{r_{ds} + R_D}{1 + (g_m + g_{mb})r_{ds}} \approx \frac{1}{g_m + g_{mb}} + \frac{R_D}{(g_m + g_{mb})r_{ds}}$$